



ALL KIDS CAN PLAY

Grade **01**

ALIGNED WITH INTERNATIONAL STANDARDS

This book supports essential educational frameworks that shape future-ready minds:

- 21st Century Skills
- Social-Emotional Learning (SEL)
- Sustainable Development Goals (SDGs)
- Universal Design for Learning (UDL)

STEAM Scouts is a ground-breaking STEAM book series with an interdisciplinary focus designed to ignite curiosity, creativity, and critical thinking in learners worldwide. The series supports Social-Emotional Learning (SEL) by promoting core competencies like self-awareness, self-management, social awareness, relationship skills, and responsible decision-making. It also aligns with the UN Sustainable Development Goals (SDGs), fosters 21st Century Skills, and meets international educational standards to equip learners with tools for a dynamic, innovative, and impact-driven future. Furthermore, it is built on Universal Design for Learning (UDL) principles, ensuring inclusive, accessible, and engaging STEAM education for all students, regardless of learning ability.

Key Features

Standards-Aligned

- NGSS (Next Generation Science Standards) – Authentic scientific inquiry & engineering design.
- CCSS (Common Core State Standards) – Rigorous literacy & math integration.
- Cambridge & IB (PYP/MYP/DP) – Global perspectives & interdisciplinary learning.
- ISTE (Technology Standards) – Digital literacy & computational thinking.
- STEL (Standards for Technological & Engineering Literacy) – Real-world problem-solving.

Social-Emotional Learning (SEL) and Global Citizenship

- Each book fosters collaboration, resilience, leadership, and ethical decision-making while connecting to multiple UN Sustainable Development Goals (SDGs) - such as Quality Education (SDG 4), Clean Energy (SDG 7), Climate Action (SDG 13), and Innovation (SDG 9) - empowering students to think globally, encourages ethical innovation & community impact.
- Differentiated challenges to meet diverse learning needs and skill levels.
- Visual cues and infographics to enhance understanding and engagement.
- Guided instruction to support learners through new or complex concepts.
- Scaffolded activities that build skills progressively and promote learner independence.

Why Choose This Series?

Future-Ready Skills – Prepares learners with essential skills like creativity, collaboration, and critical thinking for future careers and global challenges.

Teacher & Student-Friendly – Includes engaging lesson hooks, discussion prompts, guided instruction, and hands-on tasks for seamless learning.

- **SDG-Themed Modules** – Each module links to 2+ UN SDGs, blending academics with global purpose and ethical action.
- **Inclusive & Engaging** – Built on UDL principles to ensure accessibility through differentiated pathways, visual supports, and hands-on learning.

Perfect For:

- K-12 Schools adopting IB, Cambridge, NGSS, or CCSS
- STEM/STEAM Coordinators seeking industry-relevant curricula
- Homeschoolers and Educators wanting SDG-aligned, future-focused content
- Educational Institutions committed to sustainability and inclusion

Theme

03

ALL KIDS CAN PLAY

Each week has a double session, total time of session 80-90 mins



Week	Unit Name	Lesson	Page
Week 1	Unit 1: Spark The Challenge!	Lesson 1: Play for Everyone!	2-17
		Lesson 2: Turning Ideas into Action.	18-30
Week 2		Lesson 3: Designing an Inclusive Playground.	31-57
Week 3		Lesson 4: Ramp up the Fun! DIY Spiral Park.	58-74
Week 4		Lesson 5: Spinning into Inclusion.	75-91
Week 5	Unit 2: Innovators in Action.	Lesson 1: Up & Down for Everyone!	92-109
Week 6		Lesson 2: Light-Up Swings for Everyone!	110-127
Week 7	Unit 3: Beyond Imagination.	Lesson 1: Beep & Go! Guiding the Way	128-145
Week 8		Lesson 2: Find Your Way!	146-165

About The Theme

Every child deserves a fun, safe place to play—regardless of their abilities. Yet many playgrounds still fall short of being truly inclusive. In this theme, students will step into the roles of young engineers, coders, and designers as they imagine and create a smarter, more accessible playground where everyone is welcome.

Using the power of technology, coding, and AI, students will explore how intelligent design can remove barriers and support all children. They will build simple electric circuits and integrate features like sound alerts, and moving components—enhanced with basic programming and AI-driven logic to adapt to different needs. These hands-on activities will teach students how circuits, motors, and sensors work together and how coding can bring their designs to life.

Throughout the project, students will dive into foundational coding concepts such as sequencing, algorithms, and conditional statements, learning how to give precise instructions to control devices and respond to real-world challenges. They'll even begin to explore how AI can recognize patterns and help create more responsive, inclusive technologies.

This project connects directly to Sustainable Development Goals 10 (Reduced Inequalities) and 11 (Sustainable Cities and Communities), inviting students to consider how empathy, fairness, and smart technologies can transform public spaces. By the end of the theme, students will have not only designed and built a model of an inclusive, tech-enabled playground—but also discovered how coding and AI can help build a more inclusive future for everyone.

HOW TO USE THIS BOOK?

1

Ready, Set, Learn!

Where we merge your existing knowledge with exciting new discoveries.

2

Inquiry-Based Learning!

Let's Think!

In this section, students will brainstorm and explore possible solutions to specific problems. This encourages critical thinking and creativity as they consider various approaches.

Explorer's Toolkit

Here, students will dive deeper into the topic. They will engage with curated videos, articles, educational games, and hands-on experiments to broaden their understanding and spark curiosity.

DIY Lab

It's part of the explore, students will undertake hands-on projects where they will design, build, and test their own creations. They will apply scientific principles, use technology and engineering skills, explore artistic concepts, and utilize mathematical reasoning.

Screen Lessons

Students will watch videos that explain key concepts in an engaging and accessible way.

3

Problem-Based Learning:

- **Introducing the problem:**
Students will join a journey to tackle real-world environmental challenges. Through hands-on activities and engaging tasks, they will apply their knowledge and skills to discover innovative solutions.
- **Apply the Engineering Design Process (EDP):**
Students will use a structured approach to define problems, conduct research, generate ideas, prototype solutions, test, evaluate, and refine their designs.
- **The Scientific Method:**
Students will follow a systematic process of inquiry—making observations, forming hypotheses, conducting experiments, analyzing data, and drawing conclusions—to develop a deeper understanding of scientific concepts and problem-solving. In doing so, they will incorporate the CER framework to present their scientific claims, provide evidence from their experiments, and explain their reasoning behind the conclusions they draw.

4

Project-Based Learning:

PBL immerses students in real-world challenges through hands-on projects. They build critical thinking, creativity, collaboration, and communication skills while designing solutions and taking ownership of their learning.

5

Story

Join Adam and Laila, two fearless STEAM Scouts, on adventures full of discovery and creativity! From building inventions to solving real-world challenges, they'll inspire you to think, experiment, and innovate. Step into the world of science, technology, engineering, arts, and math—where every challenge becomes an adventure!

Story Reflection

In this section, students will reflect on the story's key themes, characters, and challenges.

6

Assessment Exercises:

These are divided into three categories to ensure comprehensive learning and mastery.

Apprentice:

Developing understanding and can apply skills with guidance.

Builder:

Demonstrates solid understanding and can apply skills independently.

Mastermind:

Demonstrates deep understanding and can apply skills creatively and strategically.

7

Career Readiness

This section equips students with essential skills for future careers by fostering problem-solving, teamwork, and adaptability. Through hands-on projects, real-world applications, and critical thinking challenges, students develop the confidence and competencies needed for success in the ever-evolving job market.

8

Mission Innovate

Now, it's your turn! In Mission Innovate; students will apply their creativity, problem-solving skills, and knowledge to develop their own competition projects. This challenge encourages independent thinking, collaboration, and real-world innovation, empowering students to become the next generation of creators and problem-solvers.

Now I Can

This section is designed to help you measure your progress and achievements. Enjoy your exploration and learning journey through this STEAM book!

My Journal

In this part students will write their own story while they are creating the project.

By the end of this theme, students will be able to:

- Talk about how some kids might need help playing and how to make play fair for everyone.
- Show kindness by thinking about what others need to have fun and feel included.
- Explore how shapes, colors, and natural materials can make playgrounds more fun for everyone.
- Measure and compare spaces and materials to see what works best for safe and fair play.
- Follow the steps of the Engineering Design Process to solve a problem by imagining, planning, creating, testing, and improving their ideas.
- Measure and compare the length, width, and height of playground parts to make sure they fit and are safe for everyone.
- Draw and sort their ideas using pictures, charts, and diagrams.
- Build with 2D and 3D shapes and use patterns and bright colors to help others play.
- Use ramps, shapes, and colors to build fun and safe play areas.
- Try out different materials to see which ones are strong and safe.
- Make simple circuits to light things up or make sounds.
- Use m.Block to help a character move through a fun maze.

Unit

1

Lesson

1

Spark The Challenge!

Play for Everyone!

Time Frame: [70 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Talk about ways kids play differently and why some children may need help to play.
- Share ideas on how to make play fair and fun for everyone.
- Show kindness by thinking about what other kids might need to feel happy and included.
- Name one or two things that make a playground good for all children.
- Look at shapes, colors, and natural things like sand and plants to see how they make playgrounds fun for everyone.
- Measure and compare spaces and materials to find out which ones are safe and good for play.

Challenge questions of the lesson

1. What might make it hard for some friends to play on a playground?
2. How can we help everyone feel happy and included when they play?
3. If you were building a new playground, what's one thing you would add to make sure all kids can play?

SEL Relation

- Empathy (Thinking about others): We will think about how our friends might feel if they can't play easily on the playground.
- Inclusion (Making everyone feel welcome): We will learn why it's important for everyone to be able to join in and have fun together.
- Fairness (Making things right for everyone): We will talk about how playgrounds can be fair so that all kids have a chance to play.
- Perspective-Taking (Seeing things from another's view): We will try to imagine what it would be like if we couldn't use some of the playground equipment.

Engage



Teacher Role:

- **Facilitator:** Guide the discussion, now referencing specific events and dialogue from the story. Encourage students to connect the illustrations with the narrative.
- **Observer:** Pay attention to students' understanding of the characters' feelings and the problem presented in the story.
- **Clarifier:** Help students understand any vocabulary or nuances in the dialogue.

Expected From Students:

- **Comprehension:** Students should be able to understand the events and dialogue in the provided story.
- **Empathy:** Students should be able to identify Khaled's emotions based on his words and the situation described.
- **Problem Identification:** Students should be able to pinpoint the specific problem Khaled faces on the playground.
- **Connection to Visuals:** Students should be able to relate the illustrations to the characters and events in the story.
- **Critical Thinking:** Students should be able to justify their answers based on both the story and the visual information.

Parts of the books included



Story



1. Set the Scene with Emotion and Curiosity

- Teacher says (dramatically):

“The sun was shining and Miss Sarah’s class went on a trip to the park! Can you imagine how fun that would be? What’s your favorite thing to do at the playground?”

(Let children respond – this hooks them emotionally and personally.)

- Then:

“But... something surprising happened. Let’s listen and see...”

2. Read the Story Aloud with Expression

Use gestures, voices, and movement to bring the story alive:

- Use a sad voice for Khaled,
- An excited voice for Adam and Laila,
- Pause and show pictures or visuals of a playground (optional).

Possible Strategies for Implementation:

1. Role-Play the Story (Social-Emotional Learning)

- Have students act as Khaled, Laila, and Adam.
- Let them imagine and act out ways to help.

2. Vocabulary and Sentence Building

- Highlight and practice key words: slide, ramp, playground, sad, fix.
- Let students form simple sentences like:

“I play on the slide.”

“Khaled is sad

3. Story Retelling with Puppets or Pictures

- Give kids simple puppets or image cards and ask them to retell the story.

Story Reflection



Objective: ensures students comprehend the story and recognize how exclusive playgrounds impact the feelings of individuals with disabilities.

Answers:

1. How did Khaled feel when he saw other children playing on the slide?

- **Expected Answer:** a) Sad
- **Reasoning:** The story states Khaled "sighed" and said, "I want to... but the ramps are too steep, and I can't get on the slide." This, combined with the visual of Khaled looking at the other children, indicates feelings of sadness, disappointment, and exclusion.

2. What is the main problem that Adam and Laila notice in the playground?

- **Expected Answer:** b) Some children can't use all the equipment.
- **Reasoning:** In the story, Khaled explicitly states the ramps are too steep for him to use the slide. Adam's reaction, "That's not fair! Everyone should be able to play!" and Laila's suggestion, "We need to fix this. What if we make a playground where everyone can play, even you?" directly highlight the problem of the playground not being accessible to all children.

Possible Strategies for implementation:

1. Guided Discussion with Visual Cues

- Use emotion cards (happy, sad, frustrated) or pictures of faces from the book to help students identify how Khaled felt.
- Ask:
 - "Which face shows how Khaled felt?"
- Then follow up:
 - "Why do you think he felt that way?"

Purpose: Enhances emotional vocabulary, fosters early inference abilities.

2. Multiple Choices with Movement

- Create interactive answer stations: each corner of the room represents an answer (A, B, C).
- Ask a question (e.g., "How did Khaled feel?") and let students walk to the corner they think is correct.
- Ask students in each corner to explain why they chose their answer (with help if needed).

Purpose: Makes thinking active and inclusive for all learners.

Let's Think!



Objective:

Students will create a model of an inclusive playground by selecting, cutting, and pasting images and/or drawing their own ideas—promoting creativity, design thinking, and empathy.

Resources to print for this activity:

<https://drive.google.com/file/d/1Uz2foUX1mGXbR-OEZm8b24Bti61UHHfT/view>



Activity Instructions

Step 1: Set the Stage

- Say: “We’re going to be playground designers today! But not just any playground—we’re building a playground where everyone can play.”
- Show the empty playground photo on the board or ask them to open it in the book.

Step 2: Activate Thinking

- Briefly review the story of Khaled and the problem in the park.
- Ask:
 - “What made it hard for Khaled to play?”
 - “What could help him and others have more fun?”

Step 3: Explore Materials

- Distribute:
 - Cut-out images (swings, ramps, sandboxes, sensory panels, etc.)
 - Crayons, markers
 - Glue sticks and safety scissors
- “You can glue pictures on the playground OR draw your own ideas! Use shapes, colors, and your imagination.”



Step 4: Design Time (Student Work Time)

- Encourage:
 - Use of both cutouts and drawings
 - Thinking about inclusion (e.g., ramps, quiet spaces, adaptive swings)
 - Collaboration if working in small groups

Step 5: Share & Reflect

Let students present their playgrounds.

- Prompt with questions:
- “What did you add for kids who use wheelchairs?”
- “Where can someone go if they want quiet?”
- “Why did you choose these colors or shapes?”



Explorer's Toolkit



Teacher Role:

- **Facilitator:** Guide students through the observation process by pointing out different playground features and encouraging them to use the checklist.
- **Questioner:** Ask probing questions to encourage deeper thinking about why certain features are important for inclusivity (e.g., "Why are ramps important?", "How does a quiet space help some children?").
- **Model:** If needed, model how to fill out the CER diagram with a simple example related to fairness or accessibility.
- **Support:** Offer assistance to students who are struggling with observation or articulating their claims, evidence, and reasoning.
- **Connector:** Help students connect their observations to the story and previous discussions about Khaled's experience.

Expected From Students:

- **Active Observation:** Students will carefully look at the provided playground image (or an actual playground if available) and identify the features listed in the "Playground Checklist."
- **Hands-on Participation:** Students will physically check the boxes on the checklist as they locate the features.
- **Critical Thinking:** Students will analyze whether the observed features contribute to a playground that is "for ALL kids to play, especially friends who use wheelchairs or need calm spaces."
- **CER Diagram Completion:** Students will formulate a claim about whether the playground is fair for everyone, provide specific evidence from their observations, and explain their reasoning based on the importance of inclusivity.
- **Articulation of Ideas:** Students will express their thoughts and observations using words and potentially drawings in the CER diagram.

Parts of the books included.



DIY Lab

Objective:

Students will observe and evaluate the accessibility of their school playground using a checklist, then apply the Claim-Evidence-Reasoning (CER) framework to form and express an opinion about whether the playground is fair and inclusive for all children. Through this activity, students will develop skills in critical thinking, observation, empathy, and communication while making real-world connections to inclusive design.

Materials Needed:

- The "Playground Checklist" worksheet (provided in the book).
- Pencils or pens for checking boxes.

Activity Instructions

The teacher will take the students to the school playground during the garden lesson, where they will explore the area and mark items on their checklist.

Introduce the CER Diagram: Explain that CER stands for Claim, Evidence, and Reasoning. It's a way to share their ideas clearly and support them with what they saw.

- **Formulate a Claim:** Guide students to make a claim (a statement) about whether they think the playground is fair for everyone based on their observations from the checklist. They should complete the sentence: "I think this playground is/is not fair for everyone."
- **Provide Evidence:** Next, students need to provide evidence (what they saw) to support their claim. They should look back at their completed checklist and choose specific features they observed (or didn't observe) that support their claim. They should complete the sentence: "I saw (e.g., ramps, swings for all kids)." Encourage them to be specific.
- **Explain Reasoning:** Finally, students need to explain their reasoning (why their evidence supports their claim). They should think about why those features are important for making a playground fair and welcoming to all children, connecting back to the idea that "all kids should play and feel happy." They should complete the sentence: "It matters because (e.g., all kids should play and feel happy)."

Possible Strategies for implementation:

1. Explore with Purpose (Checklist Walk)

Strategy: “See, Check, Talk”

- Students observe the playground.
- They check off items like ramps, swings, shady spots, quiet spaces, etc.
- Encourage peer discussion with a buddy:
- “What did you see? Do you think that helps everyone?” (Supports observational skills and social learning.)

2. Make a Claim

Strategy: “Thumbs & Talk”

- Use thumbs up/down/sideways for a quick check:
 - “Do you think the playground is fair for everyone?”
- Then students use sentence starters:
 - “I think this playground is/is not fair for everyone because...”
- Helps students express opinions confidently with structure.

3. Provide Evidence

Strategy: “Evidence Detective”

- Students look back at their checklist like detectives.
- Provide sentence frames:
 - “I saw ____ that helps all kids.”
 - “I didn’t see ____ which would help.”
- Use visual aids or photos from the walk for reference.
- Encourages use of specific, concrete examples from experience.

4. Explain Reasoning

Strategy: “Why Does It Matter?” Circle Time

- Sit in a circle. Teacher prompts:
 - “Why is it important that all kids can use the playground?”
- Students use:
 - “It matters because...”
 - “Kids who use wheelchairs need ____ to play.”
- Optionally, use empathy role-play:
 - “Imagine if you couldn’t use the slide—how would you feel?”
- Deepens understanding of fairness and empathy.

Explain



Teacher Role:

- **Facilitator:** Introduce the video and guide students through the pre-, during-, and post-watching activities.
- **Purpose Setter:** Clearly explain the learning objectives and what students should be looking for in the video.
- **Engagement Encourager:** Prompt students to actively observe and think critically while watching.
- **Discussion Moderator:** Lead the post-watching discussion, ensuring all students have an opportunity to share their observations and insights.
- **Concept Clarifier:** Address any questions or misunderstandings that may arise after watching the video.
- **Connection Maker:** Help students connect the information in the video to previous activities and discussions about inclusivity.

Expected From Students:

- **Active Viewing:** Students will watch the video attentively, focusing on identifying features that promote inclusive play.
- **Note-Taking (Optional but encouraged):** Students may jot down key features, ideas, or questions that arise during the video.
- **Thoughtful Participation:** Students will contribute meaningfully to the pre- and post-watching discussions, sharing their initial thoughts, observations, and reflections.
- **Critical Analysis:** Students will think critically about the information presented in the video and how it relates to creating fair and accessible playgrounds.

Parts of the books included.



Screen Lessons

Watch the video to learn about things that help all kids play at the playground.

<https://ai.invideo.io/watch/yPHVup1oPzE>

Pyramakerz*



Screen Lessons

Objective: Students will develop a deeper understanding of inclusive playground design by actively engaging with video content through anticipation, focused observation, and structured reflection. They will practice critical thinking, listening, and communication skills as they analyze features that make playgrounds accessible for all children, revise their initial assumptions, and articulate new insights about fairness and inclusion in play environments.

Video link: <https://ai.invideo.io/watch/yPHVup1oPzE>



Possible Strategies for implementation:

1. Pre-Watching: Set a Purpose

- Strategy: Anticipation Guide
 - How to Implement:
 1. Before showing the video, present students with 3-2 statements related to inclusive playgrounds (without revealing if they are true or false). Examples:
 - "Playgrounds only need swings and slides to be fun for everyone."
 - "Ramps and wide pathways can help more children enjoy a playground."
 - "Quiet spaces are not important on a playground."
 2. Ask students to individually agree or disagree with each statement and briefly jot down their reasons before watching the video.
 3. Explain that the video will help them confirm or revise their initial thoughts.

2. During Watching: Active Engagement

- Strategy: Focused Observation with Prompts
 - How to Implement:
 1. Before starting the video, provide students with 3-2 specific questions or prompts to focus on while watching. Examples:
 - "What are some different ways children with mobility challenges can play on an inclusive playground shown in the video?"
 - "What features in the video seem designed to help children who might feel overwhelmed?"
 - "What are some examples of playground equipment in the video that can be used by many different children?"
 2. Encourage students to pay close attention to these aspects as they watch. They can take brief notes if helpful.

3. Post-Watching: Discussion and Reflection

- Strategy : Think-Pair-Share (Revisited with Video Focus)
 - How to Implement:
 1. Pose a question related to the video's content (e.g., "What was one new thing you learned about inclusive playgrounds from the video?" or "How did the video change your thinking about what makes a playground fair?").
 2. Give students individual "think time."
 3. Have them pair with a partner to discuss their thoughts.
 4. Facilitate a whole-class share-out of key insights.

Elaborate and Evaluate



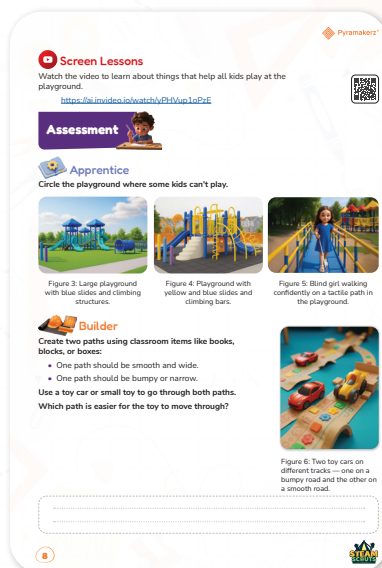
Teacher Role:

- **Clarifier:** Ensure students understand tasks such as identifying inclusive vs. non-inclusive playgrounds and building different types of paths. Emphasize key concepts like accessibility, smooth vs. bumpy, and wide vs. narrow.
- **Facilitator:** Provide necessary materials and a safe, organized space for building and exploring. Set up the activity with clear instructions and supportive structure.
- **Encourager:** Inspire creativity and experimentation. Celebrate different ideas and approaches to building and problem-solving.
- **Guider:** Prompt students to reflect on design choices using guiding questions like:
“Why do you think the toy moves easier on that path?”
“What makes this playground harder for some kids to use?”
- **Questioner:** Ask open-ended questions during activities to deepen student thinking and connect observations to real-world design principles.
- **Observer:** Watch for how students analyze images, apply inclusive design knowledge, solve problems during the building phase, and explain their thinking clearly.

Expected From Students:

- **Visual Analysis:** Examine playground images carefully and identify features that make some spaces less inclusive (e.g., no ramps, narrow entrances, unsafe surfaces).
- **Application of Knowledge:** Use prior learning about inclusive design to make informed choices and judgments.
- **Creative Construction:** Use classroom materials in inventive ways to create two types of paths — one smooth and wide, and one bumpy and narrow.
- **Model Building:** Construct the paths with attention to specific physical properties and test them with toy movement.
- **Experimentation:** Engage in hands-on testing of toy movement across both paths to gather firsthand evidence.
- **Observation:** Notice differences in how easily toys move over different surfaces and begin to draw connections to real-world accessibility.
- **Explanation:** Clearly communicate which path worked better and why, using evidence from their own models and observations.

Parts of the books included.



Apprentice

Objective: Students will analyze visual representations of playgrounds to identify features that either support or limit accessibility. They will apply their understanding of inclusive design principles to evaluate which playground may exclude some children

Answer: The playground in the middle image (with the yellow and blue equipment on dark mulch and steps leading to the slide) is likely the one where some kids can't play. The first playground appears to have ramps and accessible surfacing. The third image shows a ramp, suggesting accessibility.

Circle the playground where some kids can't play.



Possible Strategies for Implementation:

1. **Visual Scanning:** Carefully look at all parts of each playground image.
2. **Compare and Contrast:** Compare the features of the three playgrounds to identify the one that seems least accessible.



Builder

Objective: Students will explore how surface design affects movement by constructing and testing two different types of paths using classroom materials. Through hands-on experimentation, they will observe how physical characteristics like smoothness, width, and bumpiness impact the ease with which a toy can move. This activity promotes scientific inquiry, problem-solving, and early engineering thinking.

Materials Needed

- Books (various thicknesses and sizes)
- Blocks (different shapes and textures)
- Small toys (cars, figures with wheels or a flat base)
- (Optional) Other classroom items like rulers, craft sticks, or textured materials.
- A flat surface to build on (desk, table, or floor).

Activity Instructions

1. **Understand the Task:** Review the instructions for the Builder level with the students, ensuring they understand the requirements for each path (smooth/wide and bumpy/narrow).
2. **Brainstorm Materials:** Encourage students to think creatively about how they can use the available materials to create the desired path characteristics. For example:
 - Smooth and wide: Using the flat sides of large books laid side-by-side.
 - Bumpy and narrow: Using small, uneven blocks placed close together or thin round objects.
3. **Build the Paths:** Have students construct their two paths on the flat surface. Encourage them to make them distinct.
4. **Test the Toy:** Once the paths are built, have students gently push or roll their chosen toy car or small toy along each path.
5. **Observe and Compare:** Ask students to pay attention to how easily the toy moves on each path.
6. **Record Observations (Optional):** Students can draw their paths and note their observations about the toy's movement.
7. **Answer the Question:** Students should answer the question:

Which path is easier for the toy to move through?

Possible answers:

- “The smooth and wide path is easier.”
- **Reasoning:**
 - “The toy moved straight and didn’t get stuck.”
 - “There were no bumps to slow it down.”
 - “It had more space to roll.”
 - “The bumpy and narrow path is harder.” (indirect answer)
- **Reasoning:**
 - “The toy kept falling off.”
 - “It was too tight and bumpy.”
 - “It got stuck between blocks.”
 - “The toy moved faster on the smooth path.”

- **Reasoning:**
 - “It went far without help.”
 - “I didn’t have to push hard.”
 - “The toy couldn’t move well on the bumpy path.”
- **Reasoning:**
 - “It wobbled or tipped.”
 - “It didn’t roll all the way.”



Now I Can

Objective: Students will reflect on their learning by identifying and expressing what they now understand about inclusive play, empathy, design, and safe playground spaces. Through the checklist, students will demonstrate their ability to recognize different needs, share inclusive ideas, and observe how materials and space contribute to safe and fun play for all children.

Activity Instructions

Step 1: Revisit the Playground Experience

- Begin the lesson by saying:
“You’ve all done amazing work thinking about how to make a playground where everyone can play. Now, we’re going to think back on what we learned and what we can now do!”
- Display the “Now I Can” checklist and read it aloud, one item at a time.

Step 2: Student Self-Reflection

- Ask them to open the checklist and provide pencils or crayons.
- Ask students to:
 - Check off what they feel confident about.
 - Draw a picture or write one thing they learned.

“Pick one thing from the list and draw how you did it. Or write about what you learned!”

Step 3: Small Group Discussion

- In groups of 3–4, have students share one thing they checked and explain:
“What did you do that helped you learn that?”
“Why do you think that’s important for kids on the playground?”

Relation to Standards

1. NGSS (Next Generation Science Standards)

1-PS4-1

Standard: Plan and conduct investigations to provide evidence that vibrating materials can make sound and that sound can make materials vibrate.

Relation: Although this is sound-focused, the investigation and observation process aligns with how students test how materials affect movement.

K-2-ETS1-1 (Engineering Design)

Standard: Ask questions, make observations, and gather information about a situation people want to change to define a simple problem.

In this lesson:

- Students explore a real-world concept (surface design).
- They ask: “Which path is easier?”

K-2-ETS1-2

Standard: Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function.

In this lesson:

- Students build physical models (paths).
- They test and observe how design affects movement.

K-2-ETS1-3

Standard: Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs.

In this lesson:

- Students compare smooth vs. bumpy paths.
- They reflect on toy performance and explain differences.

2. ISTE Standards for Students

Empowered Learner (1a, 1c)

- Students set learning goals and use technology to reflect and demonstrate understanding.

In this lesson: Students explore how design affects function, record findings, and reflect.

Innovative Designer (4a, 4b, 4c)

- Students use a design process to identify problems, test solutions, and reflect on outcomes.

In this lesson: Building paths and testing toys models the design-test-improve cycle.